

ECEN 743: Reinforcement Learning

Assignment 2

Overview

1. You will complete this assignment in a Jupyter notebook. You can download it from our (ECEN743 GitHub repo).
 2. You have to submit an HTML report and your code to Canvas.
 3. Put all your files (HTML report and code) into a **single compressed folder** named **Last-Name_FirstName_A1.zip**.
 4. Refer to the **README** of the GitHub repository for how to generate the proper HTML report for submission and for the installation instructions.
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Learning

In this assignment, you will implement a canonical model-free reinforcement learning algorithm. *Model-free* means that the learner (you) will not have the model (transition probabilities). Instead, you will sample from the environment. You will use the frozen lake environment from Gymnasium (Link).

1. **Tabular Q-Learning:** Implement tabular Q-Learning on the FrozenLake environment. You should try to optimize the rate of convergence. *Hint: Try various functions to decay exploration and learning rate. This will help you converge in a lesser number of episodes.*
 - (a) Plot G_k , where G_k is the cumulative reward obtained in episode k . Use a sliding window averaging to obtain smooth plots.
 - (b) Plot $\|Q_k - Q^*\|$, where k is the episode number and Q^* is the optimal Q function obtained from QVI. Note that the optimal Q-value function will be given to you, and you do not need to copy Q^* from your Assignment 1 submission.
 - (c) What is the policy and Q-value function obtained at the end of the learning? Are you able to learn the optimal policy?
2. **Behavior policy:** Implement tabular Q-learning with a uniformly random policy (where each action is taken with equal probability) as the behavior policy. Compare the convergence with the ϵ -greedy exploration approach. Explain your observations and inference. Can you implement a better behavior policy and show its effectiveness?
3. **TD-Learning:** Consider the following policies: (i) the optimal policy obtained from QVI, and (ii) a uniformly random policy where each action is taken with equal probability. Learn the value of these policies using:
 - (a) Monte Carlo (MC) Learning
 - (b) Temporal Difference (TD) Learning
 - (c) What are the trade-offs between MC vs TD?